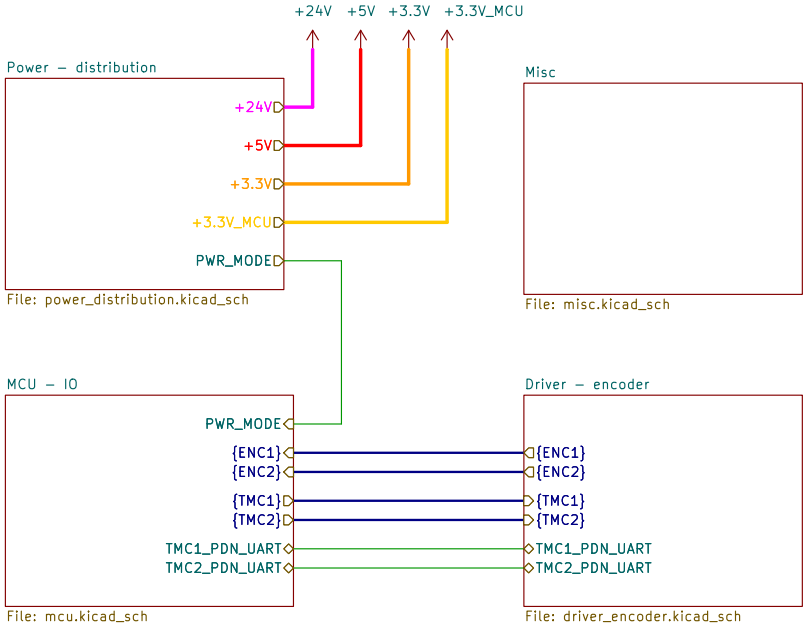
Dual Stepper Driver Evaluation Board

2026-10-22
Rev 1.0



To-Do:

- cube mx instellen: timers, GPIO, ...
- cube mx controleren

- BOM

During PCB design phase:
- HSE oscilator
- simulaties

Final design:
- should split IO in seperate sheet

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug design notes.

ELECTRICAL CHARACTERISTICS:
Voltage supplied,
Current drawn, ...

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

Sheet: /
File: stepper_motor_eval_board_v1.0.kicad_sch

Title: Stepper Motor Evaluation Board

Size: A4	Date: 2026-10-23	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/5

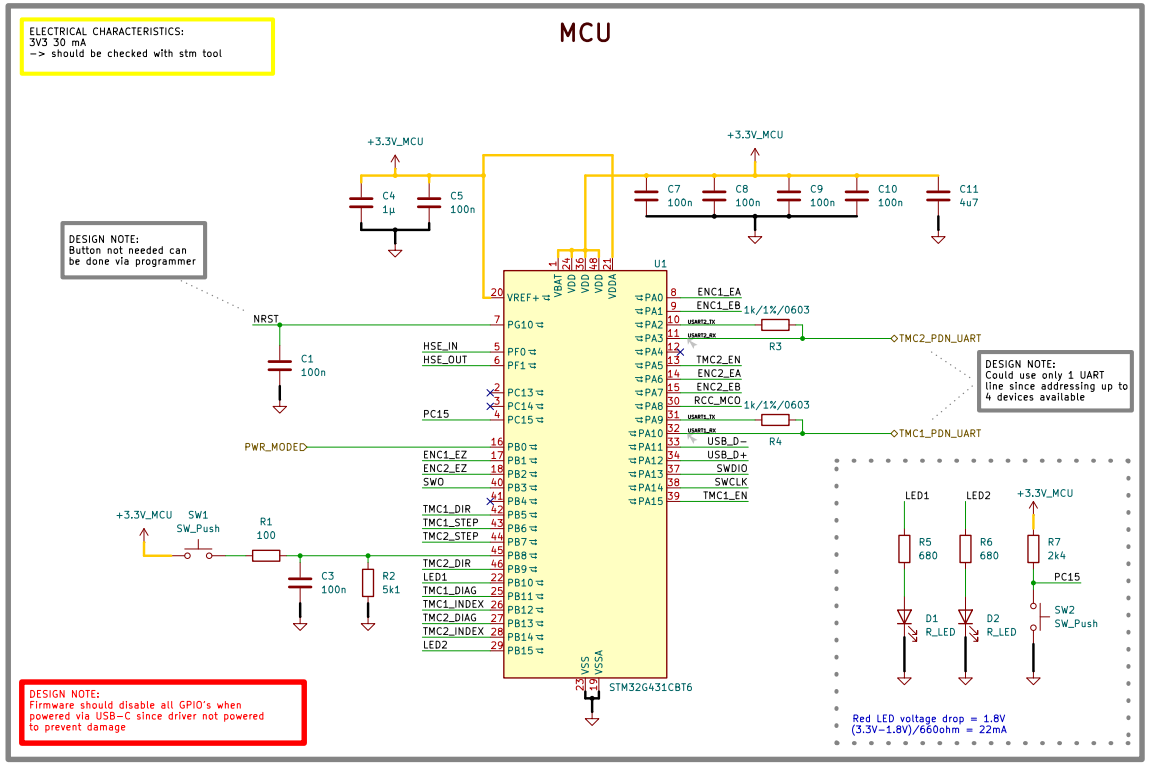
MCU

ELECTRICAL CHARACTERISTICS:
3V3 30 mA
-> should be checked with stm tool

DESIGN NOTE:
Button not needed can be done via programmer

DESIGN NOTE:
Could use only 1 UART line since addressing up to 4 devices available

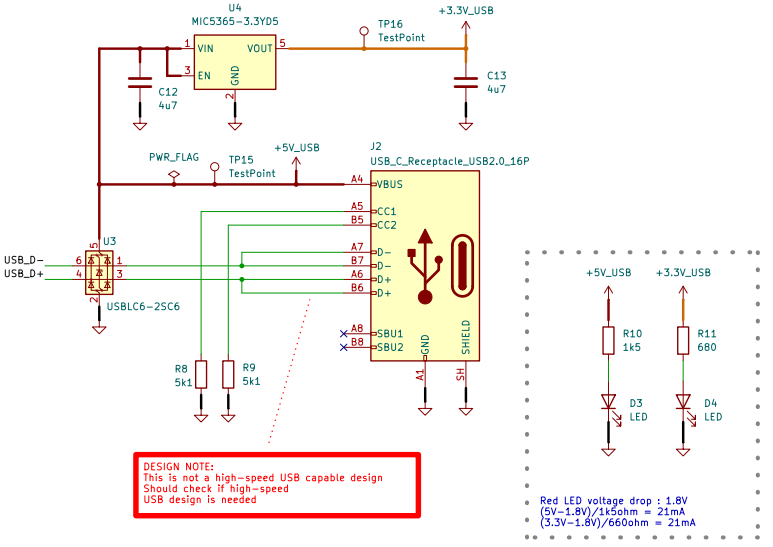
DESIGN NOTE:
Firmware should disable all GPIO's when powered via USB-C since driver not powered to prevent damage



USB

DESIGN NOTE:
USB not isolated could be in final design

DESIGN NOTE:
This is not a high-speed USB capable design
Should check if high-speed USB design is needed



HSE Oscilator

DESIGN NOTE:
No need of LSE oscilator since no sleep activity

DESIGN NOTE:
HSE oscilator since space and BOM allows it
Better accuracy timing for everything including UART and USB

DESIGN NOTE:
There are no constraints on crystal size, then it is recommended to choose a crystal with the smallest possible ESR.

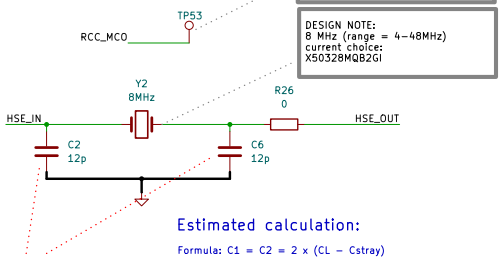
DESIGN NOTE:
If there are no constraints on power consumption, it is recommended to choose crystals with high load capacitance.

DESIGN NOTE:
Allows to measure Microcontroller Clock Output pin Without adding capacitance

DESIGN NOTE:
8 MHz (range = 4-48MHz)
current choice:
X50328MQB2GI

DESIGN NOTE:
Capacitors values should still be calculated depending on PCB traces etc.

Use 0402 footprint for easy replacement of caps when testing



Estimated calculation:

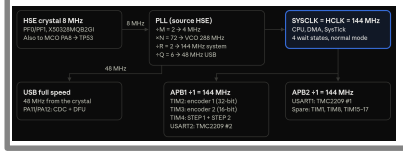
Formula: C1 = C2 = 2 x (CL - Cstray)

with Cstray estimated = 10pF p.100 datasheet MCU
C1 and C2 should be between 5pF-20pf

X50328MQB2GI:
CL = 16pF -> so that C1, C2 are in middle of allowed range
ESR = 80ohm
Tolerance = +-10ppm
Drive level = 100µW or higher

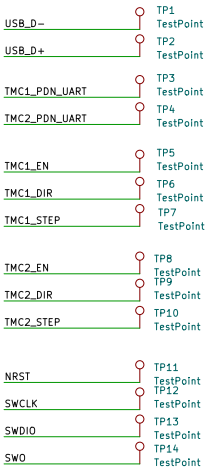
C1 = C2 = 12pF (estimate) -> good in middle of range

Clock tree

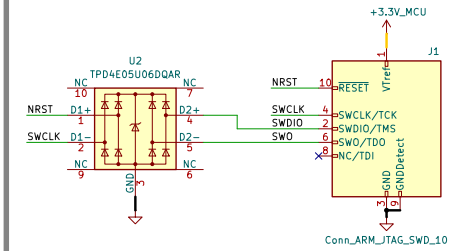


LAYOUT NOTE:
- place crystal and caps very close to OSC pins
- keep traces short and symmetrical
- avoid noisy digital routing near the oscillator loop
- avoid excessive vias in crystal traces
- respect the datasheet's ESR and drive recommendations

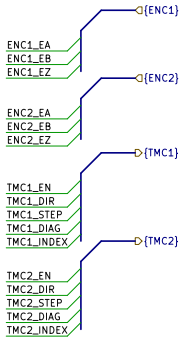
Testing points



SWD (Debug) Port



Interconnections



Sheet: /MCU - IO/
File: mcu.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 2/5

Stepper motor drivers

ELECTRICAL CHARACTERISTICS:
Driver current depending on motor: 1A/phase with 2ohm phase resistance
Pmotor_coil = (1A)² x (2ohm) = 2W
Motor has two coils so : 2W x 2 = 4W per motor
Idriver = P/U = 4W / 24V = 0.167A
Two drivers in total so : 0.167A x 2 = 0.33A

Two drivers = 24V 0.5A

One driver IO = 3V3 30µA
Two driver IO = 3V3 60µA

DESIGN NOTE:
Should still check layout example p82

DESIGN NOTE:
Do not power MCU output pin when IC not powered= when MCU only powered by USB

DESIGN NOTE:
Low ESR type

LAYOUT NOTE:
Place near IC with short path to die pad

DESIGN NOTE:
Chopper mode selection done via UART

DESIGN NOTE:
Diagnostic and StallGuard output. Hi level upon stall detection or driver error. Diagnostic done via UART

DESIGN NOTE:
In combination with a mechanical home switch, a more precise homing is enabled. see p.8 and p66

DESIGN NOTE:
Oscilloscope testpoints for voltage measure of stepper phases

DESIGN NOTE:
Oscilloscope testpoints for current sense through phases of stepper

LAYOUT NOTE:
Connect directly to GND plane and use low inductivity SMD type

DESIGN NOTE:
p.52 current sense resistor of 220mOhm for 1A motor

DESIGN NOTE:
Do not hot plug motor connector

Red LED voltage drop = 1.8V
(3.3V−1.8V)/660ohm = 22mA

Encoder – RS422 differential signals to logic signals

ELECTRICAL CHARACTERISTICS
1 AM26LV32E :
3V3 17mA max supply current
2 AM26LV32E :
3V3 17mA x 2 = 34mA

ELECTRICAL CHARACTERISTICS:
1 Encoder :
5V 50mA estimated, to be measured
2 Encoders :
5V 50mA x 2 = 100mA

LAYOUT NOTE:
Should check to add termination resistance based on simulation

AM26LV32EIDRG4

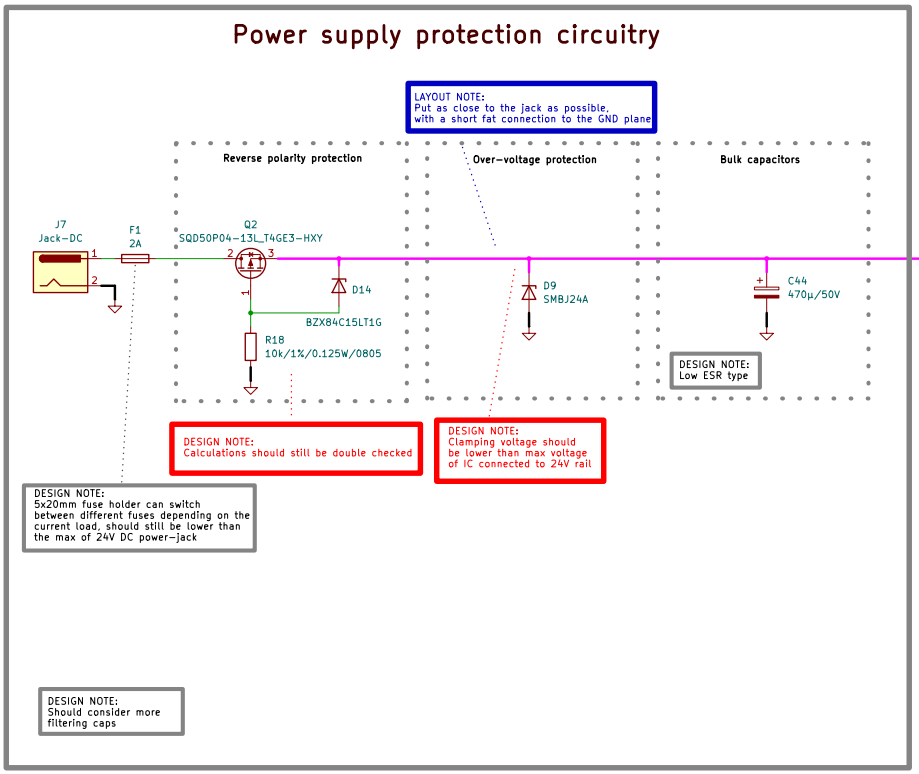
TPD6E05U06RVZR

Sheet: /Driver – encoder/
File: driver_encoder.kicad_sch

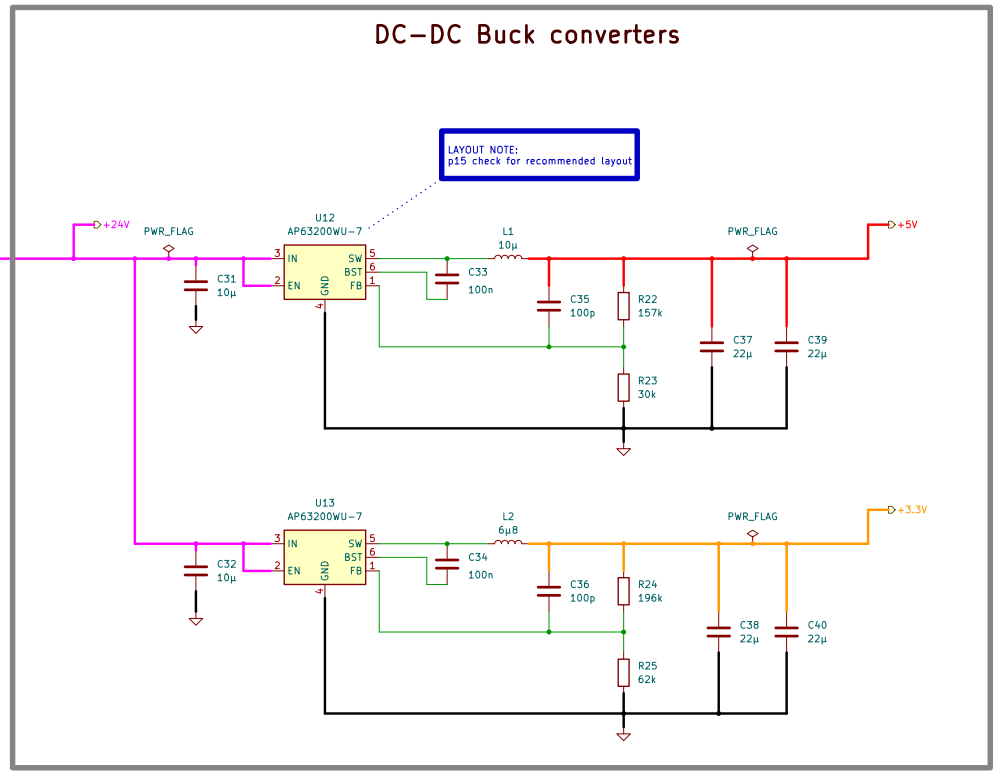
Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 3/5

Power supply protection circuitry



DC-DC Buck converters



Current consumption

Voltage rail	Components	Estimated current draw	Dimensioned current draw
24V	2x motor (driver)	0.5A	2A
5V	2x encoders	0.1A	1A
3V3	2x driver IC, RS422 signal converter	60µA + 34mA = 0.03406A	0.5A
5V_USB	MCU	30mA	0.5A

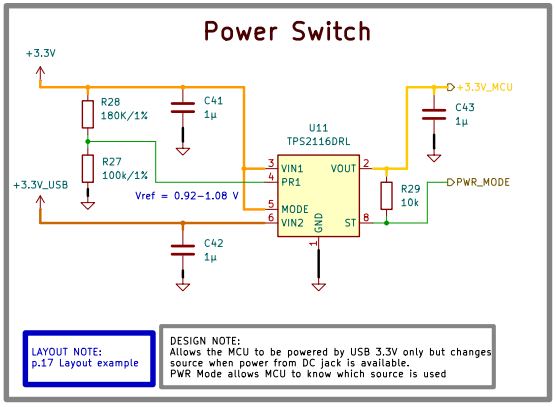
Component selector DC-DC buck converter

Table 1 shows a list of recommended component selections for common output voltages for the AP63200 and AP63201 referencing Figure 20.

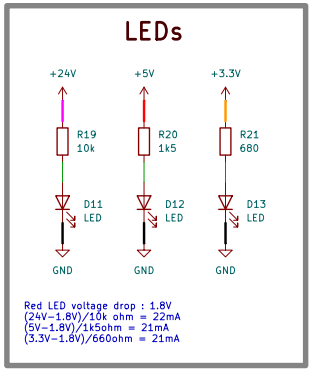
AP63200/AP63201							
Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C3 (nF)	C4 (pF)
1.2	30.9	62	2.2	10	2 x 22	100	100
1.5	54.2	62	2.2	10	2 x 22	100	100
1.8	77.5	62	3.3	10	2 x 22	100	100
2.5	131	62	3.3	10	2 x 22	100	100
3.3	196	62	6.8	10	2 x 22	100	100
5	157	30	10	10	2 x 22	100	100
12	249	18	10	10	2 x 22	100	56

Table 1. Recommended Component Selections for AP63200/AP63201

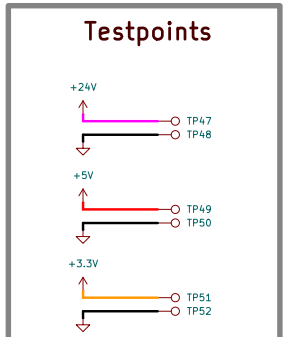
Power Switch



LEDs



Testpoints

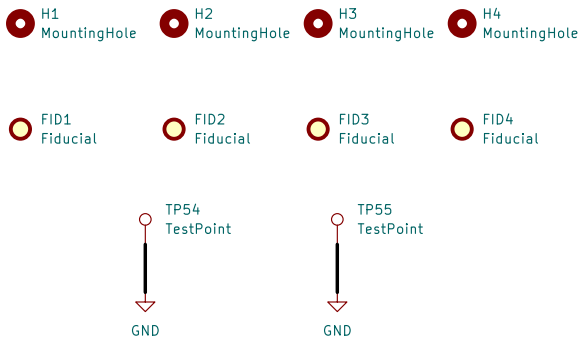


Sheet: /Power – distribution/
File: power_distribution.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 4/5

Misc



Sheet: /Misc/ File: misc.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/5